

The quantum Hall effect

Landau levels, quantized transport, and fractional charge

The quantum Hall effect connects a measurable transport coefficient to the structure of a many-electron state. The integer effect can arise without electron interactions; fractional states require them. This note follows the common starting point and then separates the two mechanisms.

1 Landau levels and their degeneracy

Consider spin-polarized electrons of charge $-e$, with $e > 0$, effective mass m_* , and a uniform perpendicular field of magnitude $B > 0$. For a parabolic band,

$$H = \frac{(\mathbf{p} + e\mathbf{A})^2}{2m_*}, \quad E_n = \hbar\omega_c \left(n + \frac{1}{2}\right), \quad \omega_c = \frac{eB}{m_*}, \quad n = 0, 1, \dots \quad (1)$$

Here $\nabla \times \mathbf{A}$ is the magnetic field; reversing its direction reverses the Hall response but not these energies. The magnetic length and flux quantum are

$$\ell_B = \sqrt{\frac{\hbar}{eB}}, \quad \Phi_0 = \frac{h}{e}. \quad (2)$$

For area A , a Landau level contains $N_\Phi = BA/\Phi_0 = A/(2\pi\ell_B^2)$ orbitals, up to boundary corrections. One way to count them is to use Landau gauge: allowed momenta are spaced by $2\pi/L_y$, while the corresponding guiding centers are spaced by $2\pi\ell_B^2/L_y$ across a sample of width L_x .

With N electrons and density $n_e = N/A$, define the filling factor

$$\nu = \frac{N}{N_\Phi} = 2\pi\ell_B^2 n_e. \quad (3)$$

Integer filling fills an integer number of levels in the clean, noninteracting, spin-resolved problem. Zeeman and valley degeneracies must be treated separately when applying this count to a material.

2 What makes a plateau robust?

For p filled levels, the ideal low-temperature response is $|\sigma_{xy}| = pe^2/h$ and $\sigma_{xx} = 0$. The sign depends on field and tensor conventions. More generally, for a periodic noninteracting two-dimensional insulator,

$$\sigma_{xy} = \frac{e^2}{h} C, \quad C = \frac{1}{2\pi} \sum_{n \in \text{occ}} \int_{\text{BZ}} \Omega_n d^2k \in \mathbb{Z}, \quad (4)$$

where Ω_n is the Berry curvature with the matching orientation convention. This is the band-topological expression for the integer response.

Disorder can localize bulk states between extended states, allowing density to change over a range without changing the Hall plateau. Thus a mobility gap, rather than a perfectly empty spectral gap, can suffice. The clean Bloch-band formula is not itself a derivation of localization or of a finite plateau width.

3 Why a fractional plateau is a many-body problem

At a partially filled, dispersionless Landau level, kinetic energy no longer selects a unique state. Interactions act within the macroscopically degenerate subspace. A model wavefunction for an incompressible state at $\nu = 1/m$, with m a positive odd integer, is the Laughlin state [2]:

$$\Psi_m(z_1, \dots, z_N) = \mathcal{N} \prod_{i < j} (z_i - z_j)^m \exp \left[- \sum_i \frac{|z_i|^2}{4\ell_B^2} \right]. \quad (5)$$

Here $z_i = x_i + iy_i$, $\mathcal{N}$ normalizes the state, and the field orientation is chosen so the lowest-level orbitals are holomorphic. The integer m is unrelated to the effective mass m_* . Odd m makes the state antisymmetric under exchange; the repeated zero suppresses close particle encounters. Counting the largest polynomial degree gives $N_\Phi \simeq mN$, hence $\nu \rightarrow 1/m$ at large N .

A quasihole at η is represented by

$$\Psi_{\text{qh}} = \prod_i (z_i - \eta) \Psi_m. \quad (6)$$

For a gapped, screened Laughlin liquid, its integrated electron deficit corresponds to charge $+e/m$. This is a collective excitation of the correlated state. It does not split an individual electron into isolated pieces.

4 Flux insertion and an effective response

On an annulus, adiabatically insert one flux quantum through the hole. Faraday's law produces an azimuthal electric field; the Hall current transfers charge between the boundaries. Integrating the radial current gives, in magnitude,

$$|\Delta Q| = |\sigma_{xy}| \Phi_0 = \nu e. \quad (7)$$

This gives one electron per filled integer level, or e/m in a Laughlin state. The argument assumes an appropriate adiabatic evolution that keeps bulk excitations suppressed.

The same low-energy response can be encoded by a compact emergent gauge field a . In units $e = \hbar = 1$, with external electromagnetic connection A , choose

$$S[a, A] = -\frac{m}{4\pi} \int a \wedge da + \frac{1}{2\pi} \int A \wedge da. \quad (8)$$

Locally, the a equation gives $m da = dA$. Substituting the solution yields $S_{\text{eff}}[A] = (4\pi m)^{-1} \int A \wedge dA$, which reproduces $\sigma_{xy} = e^2/(mh)$ in this orientation. This local response calculation leaves global sectors and boundary degrees of freedom to be handled separately.

Distinction to remember. An integer Hall state is characterized by a quantized band response; a fractional Hall state additionally has intrinsic topological order, including fractional excitations and topology-dependent ground-state degeneracy. A measured plateau is a starting point for identifying the state, not a complete specification of all its quasiparticles.

References

- [1] D. Tong, *Lectures on the Quantum Hall Effect* (2016). [arXiv:1606.06687](https://arxiv.org/abs/1606.06687).
- [2] R. B. Laughlin, "Anomalous Quantum Hall Effect: An Incompressible Quantum Fluid with Fractionally Charged Excitations," *Phys. Rev. Lett.* **50**, 1395 (1983). [doi:10.1103/PhysRevLett.50.1395](https://doi.org/10.1103/PhysRevLett.50.1395).